

Fig. 2: An actual-size, single-side PCB for the LED stroboscope

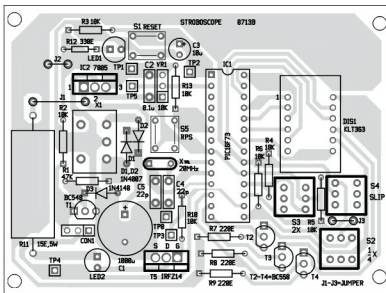


Fig. 3: Component layout for the PCB

The time period of pulses varies from 0.5 ms to 100 ms, which is arranged in two steps: when switch S2 is closed, port pin RC4 goes low and the displayed pulse rate is one-fourth the actual value. Then, if the display reads 20 ms, the actual pulse rate is 80 ms. Similarly, if S3 is closed after freezing the object, the displayed pulse rate will be double the actual value.

Switch S4 is used for slip measure-

ment of induction motors. When it holds port pin RC6 to a high logic, the display becomes inactive and pulses are derived as per the input square wave fed to port pin RC5. This square wave is generated from the AC secondary voltage of transformer X1 using transistor T1 with frequency equal to that of mains supply. Therefore by this light pulse, any AC motor will appear to be stationary if it runs exactly at the synchronous speeds. Due to this, the motor shaft appears to move in the opposite direction slowly. Since the movement is very slow, this can be counted

with a watch. With different loads on the shaft of the induction motor, the slip varies. External pulse can also be fed to port pin RC5 with suitable jumper setting of CON1.

Switch S5 is interfaced with port pin RA4. When it is pressed, the display directly shows revolutions per second on display DIS1. Otherwise, it shows the time period of pulses.

A 20MHz crystal (X_{TAL}) along with

PARTS LIST

Semiconductors:

- IC1 - PIC16F73 microcontroller
- IC2 - 7805, 5V regulator
- T1 - BC548 npn transistor
- T2-T4 - BC558 pnp transistor
- T5 - IRLF214 n-channel MOSFET
- D1-D2 - 1N4007 rectifier diode
- D3 - 1N4148 signal diode
- DIS1 - KLT363 3-digit common-anode 7-segment display
- LED1 - 5mm LED
- LED2 - 5W high-power LED

Resistors (all 1/4-watt, 45% carbon, unless stated otherwise):

- R1 - 47-kilo-ohm
- R2-R6, R10, R13 - 10-kilo-ohm
- R7-R9 - 220-ohm
- R12 - 330-ohm
- R11 - 15-ohm, 5W
- VR1 - 10-kilo-ohm multiturn trim potentiometer

Capacitors:

- C1 - 1000µF, 35V electrolytic
- C2 - 0.1µF ceramic disk
- C3 - 10µF, 16V electrolytic
- C4, C5 - 22pF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 12V-0-12V, 2A secondary transformer
- X_{TAL} - 20MHz crystal
- S1, S5 - Push-to-on tactile switch
- S2-S4 - On/off toggle switch
- CON1 - 3-pin berg strip male connector

two 22pF capacitors provides the basic clock frequency to the microcontroller. Resistor R3 and capacitor C3 are used to provide power-on reset to the microcontroller. Switch S1 is used for manual reset.

To derive the power supply for the circuit, the 230V AC mains supply is stepped down by transformer X1 to deliver a secondary output of 12 V-0-12 V, 2 A. The transformer output is rectified by a full-wave rectifier comprising diodes D1 and D2, filtered by capacitor C1 and regulated by IC 7805 (IC2). Capacitor C2 bypasses ripples present in the regulated supply. LED1 acts as the power indicator and resistor R12 limits the current through LED1.

Construction and testing

An actual-size, single-side PCB for the microcontroller-based stroboscope

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